

Chapter 94 — Upper Urinary Tract Trauma

RENAL INJURIES

- The **kidney is the most commonly injured urological organ** from external trauma.
- Radiographic staging, clinical risk stratification and wider access to **angioembolisation (AE)** have **raised renal preservation and reduced unnecessary surgery**.
- Most **blunt and selected penetrating injuries can be managed nonoperatively**; failure of nonoperative management is often **salvaged by selective AE, endourological or percutaneous intervention**.
- **Absolute indications for immediate intervention: haemodynamic instability with no or only transient response to resuscitation, or a pulsatile or expanding retroperitoneal haematoma at laparotomy.**

Epidemiology

- Kidney injured in **up to 5% of all trauma**; accounts for **about a quarter of solid-organ injuries**.
- **Blunt trauma causes 80–90% of renal injuries in the United States**. Mechanism: **direct transfer of kinetic energy and rapid deceleration**.
- Blunt mechanisms, commonest first: **motor vehicle collision (about two-thirds)** → falls from height → sports → pedestrian → assault.
- **Firearms cause about two-thirds of penetrating renal injuries**.
- **Young men predominate** (about three-quarters male; mean age about 30 years).
- **Isolated renal injury is the minority after blunt trauma**; **liver and spleen injuries** commonly coexist, and more so with higher grade. **Sports injuries are more often isolated** than other blunt mechanisms.
- Penetrating injury: associated injuries depend on **trajectory and the number and type of projectiles**.
- **Most injuries are American Association for the Surgery of Trauma (AAST) grade I–III** and are managed conservatively without significant long-term consequence. Approximate grade distribution: **I ~25%, II ~30%, III ~23%, IV ~16%, V ~6%**.
- **Greater perirenal fat thickness may protect** against high-grade blunt injury.
- National Trauma Data Bank (NTDB) data show **falling nephrectomy rates for high-grade injury**, irrespective of trauma-centre level.
- **Renal vascular injury is very rare**: blunt main renal artery injury about 0.05%; **renal vein injury rarer still** (about 0.04% of blunt trauma). Vein injuries: mostly **intimal injury with thrombus or pseudoaneurysm, usually with parenchymal injury**, and in the reported series **all managed conservatively**.

Presentation and History

- **Mechanism of injury is the single most important history** in blunt injury: vehicle speed and fall height indicate **deceleration**.
- Deceleration injures the kidney at its **retroperitoneal fixation points — renal hilum and ureteropelvic junction (UPJ)**: **renal artery thrombosis, renal vein disruption, pedicle avulsion, UPJ disruption**.
- Collision pattern: typically the **unrestrained driver** striking the steering wheel, or **side impact with door intrusion of about 30 cm or more**. First-responder detail raises suspicion.
- **Penetrating injury** (gunshot, stab): kidney involved in **about 10–20% of penetrating abdominal injuries**, mostly with other abdominal injuries. Compared with blunt injury: **more persistent bleeding, more renorrhaphy and nephrectomy, more complications under nonoperative management**.

- **Civilian gunshot wounds are usually low velocity** and are less destructive **unless they cross the hilum or collecting system**. High velocity (>2000 ft/s, >610 m/s) → **blast effect and cavitation** → **delayed tissue necrosis**.
- Stab wounds to the **upper abdomen, flank or lower chest** → suspect renal injury. **Anterior axillary line wounds** reach the **hilum and pedicle**; **posterior axillary line wounds** cause mainly **parenchymal injury**. If the weapon is recovered, **note the blade length**.
- Examination: full survey; resuscitation according to trauma protocols; **cervical spine immobilisation in blunt trauma until cleared radiographically**.
- Signs of renal injury: **flank ecchymosis, abdominal or flank tenderness, rib fractures, significant flank blow, low thoracic or flank penetrating wound**. **Ipsilateral fractures of ribs 9–12** are the most strongly associated. Consider **liver, bowel and spleen injury**.
- **Gunshot entry wounds understate internal damage** (blast effect; trajectory altered by tissue and bone; fragments act as secondary missiles). **Mark entry and exit sites with a metallic marker** on plain films to define trajectory.

Haematuria

- **Gross haematuria, or microhaematuria (>5 red blood cells per high-power field [RBC/hpf]) with shock (systolic blood pressure [SBP] <90 mm Hg)**, indicates significant injury, especially with deceleration or penetrating mechanisms.
- **Degree of haematuria does not correlate with severity**, and **its absence does not exclude injury**: gross haematuria occurs with contusions; haematuria is **absent in a proportion of grade IV injuries, renal gunshot wounds and about a third of blunt renovascular injuries**; about half of UPJ injuries have no haematuria.
- Blunt trauma: **microhaematuria with shock significantly raises the rate of severe injury**.
- **Recreational (mainly blunt) injuries** less often meet haematuria criteria; relying on haematuria alone would **miss about a quarter of grade II–IV injuries**.
- **Use the first urine aliquot** (catheter or void): later samples are **diluted by resuscitation diuresis** and underestimate haematuria.
- **Dipstick**: rapid, low risk, false-negative 3–10%, **sensitivity and specificity >97%**, grade correlates with RBC/hpf — but **microscopy is the gold standard**.
- **False-positive dipstick**: myoglobinuria, dehydration, exercise, menstrual blood, povidone-iodine.
- Haematuria must **never be the sole determinant**; mechanism and associated injuries count, and staging is **by computed tomography (CT)**.

Concept — *Haematuria is a trigger for imaging, not a measure of injury: the worst injuries (pedicle, UPJ) are precisely those that may not bleed into the urine, so mechanism carries equal weight.*

Classification

- **AAST Organ Injury Scale is the most widely used classification; updated 2018**.
- Grading by **contrast-enhanced CT**; validated to predict **surgical or angiographic intervention and nephrectomy**. In one multicentre study the **2018 revision did not improve prediction of bleeding interventions**.
- 2018 changes: **grades I–II essentially unchanged**; grade IV now includes **segmental vascular injury (arteriovenous fistula, pseudoaneurysm)** and renal pelvis injury or UPJ avulsion; grade V now includes a **devascularised kidney with active bleeding and a shattered kidney with loss of identifiable anatomy**.

AAST KIDNEY INJURY SCALE — 1989 (TABLE 94.1) VERSUS 2018 REVISION (TABLE 94.2)

Grade	1989 (Moore et al., 1989)	2018 revision (Kozar et al., 2018) — imaging criteria
I	Contusion (haematuria, urological studies normal); subcapsular non-expanding haematoma, no laceration	Subcapsular haematoma and/or parenchymal contusion without laceration
II	Non-expanding perirenal haematoma confined to retroperitoneum; laceration <1 cm, no extravasation	Perirenal haematoma confined to Gerota's fascia; laceration ≤1 cm, no extravasation
III	Laceration >1 cm, no collecting-system rupture or extravasation	Laceration >1 cm, no collecting-system rupture or extravasation; any injury with vascular injury or active bleeding contained within Gerota's fascia
IV	Laceration through cortex, medulla and collecting system; main renal artery or vein injury with contained haemorrhage	Laceration into collecting system with extravasation; renal pelvis laceration and/or complete ureteropelvic disruption; segmental artery or vein injury; active bleeding beyond Gerota's fascia ; segmental or complete infarction from thrombosis without active bleeding
V	Completely shattered kidney; avulsion of hilum devascularising kidney	Main renal artery or vein laceration or hilar avulsion; devascularised kidney with active bleeding ; shattered kidney with loss of identifiable anatomy
Rules	Advance one grade for bilateral injuries, up to grade III	Advance one grade for multiple injuries, up to grade III; grade by the highest grade on imaging, operation or pathology

- **What the table is for:** grading drives intervention thresholds; the 2018 version adds **operative and pathological criteria** alongside imaging.
- **Pattern:** grades I–III are defined by **depth** (1 cm is the cut-off); grades IV–V by **collecting system and vessels**.
- **Trap:** **main vessel injury with contained haemorrhage was grade IV in 1989 but is grade V in 2018**; segmental vascular injury is now grade IV. **Contained vascular injury within Gerota's fascia is now grade III.**
- Figs 94.1 and 94.2 illustrate the 1989 and 2018 grades (2018 figure labels pseudoaneurysm and arteriovenous fistula, active bleeding, urinary extravasation, segmental thrombosis).

Concept — *Depth defines the low grades; the collecting system and the vessels define the high grades, and the 2018 revision moved the vascular injuries into the grade that matches their bleeding risk.*

Indications for Renal Imaging

American Urological Association (AUA) Urotrauma Guideline (2014, amended 2020) and European Association of Urology (EAU) Urological Trauma Guidelines (2003, updated 2022) agree on imaging:

1. **Penetrating trauma with likely renal injury** (abdomen, flank, ipsilateral rib fracture, flank ecchymosis, low chest wound) **if stable enough for CT** rather than going straight to theatre or angiography.
 2. **Blunt trauma with significant acceleration/deceleration** (high-speed collision, fall from height).
 3. **Blunt trauma with gross haematuria.**
 4. **Blunt trauma with microhaematuria and hypotension** (SBP <90 mm Hg at any time during evaluation and resuscitation).
 5. **Children with shock and microhaematuria (>5 RBC/hpf).**
- Basis: the **San Francisco General Hospital prospective series (>25 years)** — gross haematuria, or microhaematuria with shock, → **CT with intravenous (IV) contrast and delayed films.**

- **Microhaematuria without shock or deceleration → observe without imaging;** significant injury is **vanishingly rare (<0.0016%)**. Image anyway if history, examination or course suggests injury.
- **Rapid deceleration** carries higher risk of **pedicle or UPJ injury**.
- **Penetrating injury with any degree of haematuria → image:** pedicle injury has occurred with near-undetectable microhaematuria.

FIG. 94.3 — SELECTIVE IMAGING AFTER ADULT BLUNT TRAUMA (MILLER AND MCANINCH, 1995)

Group	Management	Outcome
Microhaematuria, no shock	Most observed without imaging; a minority imaged	No significant renal injury missed; imaged patients almost all contusions
Gross haematuria, or microhaematuria with SBP <90 mm Hg	All imaged	Significant injuries found; a subset required renal repair

Paediatric caveats (under 18 years; otherwise evaluated as adults)

- **Higher risk of renal injury and of high-grade injury** than adults after blunt abdominal trauma: **larger relative kidney size, less perirenal fat, unossified bones, less rib cover**.
- **Children may not become hypotensive** despite major blood loss — **catecholamine surge maintains pressure until about 50% of blood volume is lost; image liberally**.
- **Congenital anomalies (hydronephrosis, UPJ obstruction)** are commoner; minor trauma may cause major injury.

IMAGING THRESHOLD — ADULT VERSUS CHILD

Adult (blunt)	Child (blunt)
Gross haematuria, or microhaematuria with SBP <90 mm Hg , or deceleration mechanism	Shock with microhaematuria — but hypotension is a late sign , so imaging is used liberally

Concept — *Imaging is triggered by mechanism plus physiology; the adult rule leans on hypotension, which is exactly the sign a child withholds until late.*

Imaging Studies

Computed tomography

- **Contrast-enhanced CT with immediate and delayed images is the gold standard:** quick, sensitive, specific; defines lacerations, extravasation, associated organ injury and **haematoma size**.
- **Absent parenchymal enhancement → arterial thrombosis or transection.**
- Phases, in sequence: **arterial/nephrographic phase (~80 s) → detects arterial extravasation → delayed (excretory) phase → detects urinary extravasation;** collecting-system injury is missed if contrast has not yet been excreted.
- Delay timing varies between centres (5–10 min); a multi-institutional study found **9 minutes optimal**, with shorter delays missing extravasation.
- **Delayed films may be omitted** if the kidneys are normal and there is **no perinephric, retroperitoneal, pelvic or perivesical fluid**. If no delayed phase was obtained and collecting-system injury is suspected, **repeat CT with a urographic phase**.
- **CT has replaced IV pyelography (IVP)** for grading. The need for delayed-phase CT in children has been questioned, but **missed extravasation in high-grade paediatric injury causes morbidity** (fever, readmission).

CT FINDINGS SUGGESTING MAJOR RENAL INJURY

Finding	Suggests
Medial haematoma	Vascular (pedicle) injury
Medial urinary extravasation	Renal pelvis or UPJ avulsion
Global absent enhancement (cortical rim sign)	Main renal artery occlusion
≥2 of: perinephric haematoma rim distance >4 cm, medial laceration, vascular contrast extravasation	Brisk active bleeding; AAST grade IV

- **Haematoma rim distance predicts intervention:** bleeding-intervention rates **rise steeply from a 4 cm to a 6 cm rim**.
- Earlier **low-risk versus high-risk grade IV** stratification: **rim distance >3.5 cm, vascular contrast extravasation, complex laceration** — **2–3 factors carry a many-fold higher risk** of surgery or AE than 0–1.
- **Active intravascular contrast extravasation** (bleeding brisk enough to show on the vascular phase) is **strongly associated with subsequent AE** (Fig. 94.7: gunshot wound → arterial blush → coil embolisation).
- **Multi-Institutional Genito-Urinary Trauma Study (MiGUTS):** externally validated **nomogram predicting bleeding interventions** (Fig. 94.8).

FIG. 94.8 — MIGUTS NOMOGRAM: PREDICTORS OF BLEEDING INTERVENTION IN HIGH-GRADE RENAL TRAUMA (KEIHANI ET AL., 2019)

Predictor	Points
Penetrating mechanism (vs blunt)	34
Vascular contrast extravasation	29 (largest single dichotomous factor)
Pararenal haematoma	20
Concomitant injuries	16.5
Hypotension/shock	16
Haematoma rim distance 0–12 cm	0–100 (continuous; the largest contributor)

- Total points map to probability of bleeding intervention. **Trap: haematoma size (rim distance) outweighs any single yes/no factor.**

CT limitation — renal vein injury

- **CT cannot adequately define venous injury:** perfusion and parenchyma look normal. **Medial haematoma with normal arterial findings suggests venous injury.**
- Blunt vein injuries are rare, mostly **pseudoaneurysm with or without intimal injury and thrombus**, and **mostly accompany grade IV–V injury**. They may **bleed (surgery or AE) or tamponade**; clinical significance of CT's insensitivity is unknown.

Intraoperative one-shot IVP

- **Limited contemporary role:** unstable patient, no prior CT, **unexpected perirenal haematoma at laparotomy**, exploration or nephrectomy contemplated.
- **Main purpose: confirm a functioning contralateral kidney** — exploration often ends in nephrectomy.
- Technique: **IV push of 2 mL/kg contrast → single film at 10 minutes**.
- Useful for **urinary extravasation**, **less reliable for parenchymal injury**. **Normal study with non-expanding, non-pulsatile haematoma → exploration may be avoided**; abnormal → further imaging or exploration.
- Film quality is limited by **renal hypoperfusion in shock**; it avoided exploration in about a third in one series.
- Alternatives: most trauma surgeons now **palpate the contralateral kidney**; or **occlude the injured-side ureter with a vessel loop and give IV methylene blue** — **blue urine in the catheter confirms a functioning contralateral kidney**.

Ultrasonography

- **Focused assessment with sonography for trauma (FAST)** is used acutely but has **poor specificity for adult renal injury**; limited by **obesity, subcutaneous air, previous surgery**.
- Children: **CT preferred**, though ultrasound performs reasonably for grade III–V injury.
- Can confirm **two kidneys** and a retroperitoneal collection, but **cannot distinguish haematoma from urine leak**.

Concept — *Every CT phase answers one question — arterial phase for bleeding, delayed phase for urine — and the high-risk signs are all medial, because the pedicle and the pelvis lie medially.*

Principles of Renal Trauma Management

- Practice continues to shift toward **renal salvage and falling nephrectomy rates**, using conservative management and/or AE.
- Urologists' involvement varies between centres; **urological involvement is associated with lower nephrectomy rates**.

Nonoperative Management

AUA Urotrauma Guideline:

1. **Haemodynamically stable patients → noninvasive management.**
2. **Parenchymal injury with urinary extravasation may initially be observed.**
 - Noninvasive management: **close observation of vital signs, serial haemoglobin/haematocrit every 4–6 hours, repeat examination.**
 - **Bed rest until gross haematuria clears is empirical; early mobilisation is safe** (renal, liver and spleen injury).
 - **Failure triggers:** haemodynamic instability; **falling haemoglobin with transfusion of 2–3 units of packed red cells; new flank pain or guarding.**
 - Urinary extravasation → intervene if **enlarging urinoma, with or without pain, fever, infection or rising white cell count.**
 - **Significant injuries (grade II–V) are only about 5% of renal trauma.**
 - **Stable, accurately staged grade I–IV and selected grade V injuries → expectant management regardless of mechanism.** Grade IV–V more often need exploration, but **even these can be managed without operation if carefully staged and selected.**
 - Nonoperative shift → **fewer unnecessary operations and nephrectomies, no longer (possibly shorter) stay, and likely higher renal preservation.**
 - Expectant management is advocated for **most stable blunt parenchymal injuries, many isolated stab wounds and selected isolated gunshot wounds.** Conservative management succeeds in **>90% of blunt injuries**, even with extravasation or devitalised tissue.
 - **AAST Genitourinary Trauma Study** (prospective, multicentre): of high-grade (III–V) injuries, **most were managed expectantly**, and a further minority by **minimally invasive intervention** (AE, stent, nephrostomy, drain).
 - Grade V: NTDB analysis — **about a third managed conservatively, with lower mortality than operative management.** Conservative management was more likely with **non-penetrating injury, fewer transfusions, younger age, no associated abdominal injury and a non-level-1 centre.** High renal salvage reported for grade IV–V.

Penetrating injury

- **Selected penetrating injuries can be managed nonoperatively.**

- Penetration often **breaches Gerota's fascia, losing the tamponade** that blunt injury usually preserves; high-grade penetrating injuries are more often operated on **because of associated intra-abdominal injury**.
- Nonoperative success is **higher for stab wounds than gunshot wounds**; contemporary series show **high nonoperative success with rare delayed nephrectomy**, though a substantial proportion of penetrating high-grade injuries still need open surgery.
- **Stab wounds posterior to the posterior axillary line** seldom injure viscera. **Isolated, stable stab injury → expectant management** with serial examination; **new tenderness or guarding → explore**.

BLUNT VERSUS PENETRATING RENAL INJURY

Blunt	Penetrating
80–90% of injuries	Firearms dominate; kidney in ~10–20% of penetrating abdominal wounds
Gerota's fascia usually intact → tamponade	Gerota's fascia breached → tamponade lost
Conservative success >90%	Conservative for selected cases; more bleeding, renorrhaphy/nephrectomy and complications
Image if gross haematuria, microhaematuria + shock, or deceleration	Image with any haematuria

Exploration at laparotomy

- **Retroperitoneal exploration is not mandatory** at laparotomy for other injuries, particularly for well-staged grade I–III injury.
- **Only absolute intraoperative indication: pulsatile, expanding retroperitoneal haematoma** (life-threatening renal artery laceration).
- **Opening Gerota's fascia without an absolute indication releases tamponade** → bleeding and uncontained urine leak → **avoidable nephrectomy**; in one penetrating series, **over half of kidneys explored with Gerota's opened were removed**.

Reimaging

- AUA: consider **routine follow-up CT at ~48 hours for grade IV–V injury with deep lacerations**, or with **clinical signs of complications** (fever, chills, flank pain, bleeding, distension).
- **No routine reimaging for uncomplicated grade I–III** — rarely changes management and not cost-effective.
- Routine reimaging of all grade IV–V is being questioned: **significant complications almost always declare themselves** (fever, flank pain, falling haematocrit, increasing haematuria). Blunt segmental vascular (grade IV) injury observed without intervention; in grade IV injury with extravasation, **routine reimaging produced more interventions than reimaging for cause, without fewer complications** — interventions largely in **asymptomatic** patients.

Treatment After Failure of Conservative Management

- **Conservative management rarely fails within the first 24 hours (about 3%). Risk factors: higher grade, nonrenal abdominal injury, penetrating mechanism.**
- Penetrating injury (Trauma Quality Improvement Program [TQIP]): a small proportion were managed expectantly; failures were **all grade III–V, mostly grade IV**, and **most failures ended in nephrectomy**. AE and nephrostomy/stent rescued most who were offered them. Nonoperative success fell with grade: **~100% grade I–II → ~96% grade III → ~85% grade IV–V**.
- **Most failures need only a ureteral stent, nephrostomy or AE.**
- Intervention rates vary with centre volume; **high-volume centres intervene least**.

- **Early discharge (<48 hours)** for highly selected blunt, non-iatrogenic grade I–III injury (French multi-institutional study; excluded haemoglobin <10, transfusion within 24 hours, associated bony or visceral injury): **reintervention at 30 days non-inferior**, shorter stay, less reimaging.
- Conservative versus aggressive surgical management: **fewer complications and transfusions, shorter intensive care unit (ICU) and hospital stay, lower mortality**.
- **Delayed bleeding after discharge is rare**, typically **within 2 weeks**, and **usually amenable to AE**; in blunt high-grade injury, delayed bleeding presented at a mean of about 9 days and was mostly embolised.

Urinary Extravasation

- **Observe initially: 76–90% heal without intervention. UPJ avulsion needs early intervention.**
- Incidence: **about 29%** in studies including grade III–IV; **about half** in grade IV–V.
- **Signs of renal pelvis/UPJ injury on delayed CT: medial extravasation, medial urinoma, absent contrast in the distal ureter → prompt intervention** (endoscopic, percutaneous or open).
- **Consider early stent and/or nephrostomy** when extravasation coexists with **obstruction**: ureteric clot, congenital or acquired UPJ obstruction, large parapelvic cyst (Fig. 94.10: parapelvic cysts delaying drainage → stent despite intact ureter on retrograde study).
- **Increasing volume of devitalised parenchyma → more intervention and surgery.**
- No devitalised tissue, UPJ injury or obstruction, and stable → **expectant management; prophylactic antibiotics are commonly given without supporting evidence.**
- **Reimage or intervene** for worsening flank pain, fever, rising white cell count, ileus or peritonitis. **Symptomatic leak or enlarging urinoma → drain** (retrograde stent, nephrostomy and/or urinoma drain) — AUA-supported.
- AUA: consider **repeat imaging 48–72 hours** after initial imaging for enlarging urinoma or continued leak; whether all extravasation needs routine reimaging is debated.
- **About 29% of patients with extravasation are stented**; the avoidable fraction is unknown. **Trauma surgeons stent more readily than urologists.**
- Stent drawbacks: **theoretical contamination of perinephric collections** (infected haematoma or urinoma), **stent symptoms** and the need for **removal follow-up**.
- Persistent extravasation in severe injury: **internal stent alone usually suffices**; adding a **urethral catheter** maximises drainage.
- **Retrograde stenting impossible** with **pelvic fracture urethral distraction, severe genital trauma, complete ureteral transection, fractures preventing lithotomy → percutaneous nephrostomy ± antegrade stent.**

PERINEPHRIC FLUID COLLECTIONS ON CT

Collection	Attenuation	Other features
Urinoma	0–20 HU (Hounsfield units)	Dependent contrast pooling on delayed phase (5–20 min)
Haematoma	>30 HU	—
Abscess	>20 HU	Rim enhancement

- Collection persisting despite stent or nephrostomy → **percutaneous drain** to promote healing and prevent or treat abscess.

Concept — *Urine leaks from the parenchyma heal because antegrade flow is preserved; anything that stops urine reaching the bladder — pelvic or UPJ avulsion, clot, UPJ obstruction, a cyst — converts an observable leak into one that must be drained.*

Interventions for Renal Trauma

- **Unstable with transient or no response and renal exsanguination → immediate intervention**, most often laparotomy and damage-control nephrectomy.
- Alternatives, depending on injury and interventional radiology availability: **temporary retroperitoneal packing with AE**, or **renorrhaphy/partial nephrectomy**.
- Renal artery injury with thrombosis: **endovascular intervention**, rarely open repair.
- Grade V (NTDB): **a minority conservative**; of those operated on, **nephrectomy predominated**, with AE a distant second and renorrhaphy rare. **Intervention more likely** with higher pulse, more transfusion, concomitant injury, penetrating mechanism, level 1 centre.

Angioembolisation

- **Renal arteriography and AE are increasingly used for high-grade injury**, blunt and penetrating — stops bleeding **without laparotomy**.
- Prerequisites: **experienced team, no delay, ability to monitor and resuscitate during transfer and in the suite**. **Hybrid trauma theatres** allow simultaneous surgery and embolisation.
- **Superselective AE** is effective and less invasive → **higher renal salvage**, avoiding exploration that would otherwise end in nephrectomy.
- Historic failure rates were wide, but **repeat embolisation is highly successful**. Initial success is high in grade IV and somewhat lower in grade V.
- **Predictors of AE failure: haemodynamic instability, grade V, urinary extravasation, gross haematuria**.
- Urologists are more willing than interventional radiologists to use AE for grade IV–V. **Instability and grade independently predict surgery**.
- **Traumatic pseudoaneurysm and arteriovenous fistula (AVF)** may appear **late**, after conservative management or intervention; **treated by initial or repeat AE with high success**.

Endovascular Interventions for Renovascular Injury

- **Main renal artery injury is rare after blunt trauma** but risks **renal failure, hypertension, renal loss and death**.
- **Endovascular stents** have treated **acute thrombosis from intimal flaps**; angioplasty and stenting reported for blunt occlusion or stenosis (occasional iatrogenic haemorrhage needing embolisation).
- Limits: **warm ischaemia intolerance** and **need for therapeutic anticoagulation** in polytrauma; longer follow-up needed.

Operative Management

INDICATIONS FOR RENAL EXPLORATION OR EXPEDITIOUS AE — ABSOLUTE VERSUS RELATIVE

Absolute	Relative
1. Haemodynamic instability with no or transient response	1. Urinary extravasation with significant devascularisation (can also be closely observed)
2. Expanding or pulsatile renal haematoma (usually renal artery laceration)	2. Renal injury with colon or pancreatic injury (can be observed after enteric repair)
3. Suspected vascular pedicle avulsion	3. Arterial thrombosis
4. UPJ avulsion	4. Urinary extravasation from parenchymal injury

- Newer data soften the relative indications: **devascularisation with urine leak does well conservatively** (segmental artery injury rarely needs intervention); **concurrent renal and colonic injury need not be operated on**.
- **Extravasation alone from a grade IV laceration resolves spontaneously in up to 90%**.

- **Extravasation with >25% nonviable parenchyma → more surgical intervention and complications (abscess) → lower threshold for surgery.**

Renal Exploration

- **Transabdominal approach** — allows inspection of all intra-abdominal organs; **nonrenal injury is present in almost all penetrating cases.** Stabilise great vessels, liver, spleen, pancreas and bowel first.
- **Retroperitoneal haematoma zones: zone I central** (diaphragmatic hiatus to aortic bifurcation, kidneys laterally); **zone II lateral** (kidneys to paracolic gutters); **zone III pelvis.**
- **Principle: isolate the renal vessels before opening Gerota's fascia,** to allow immediate occlusion if massive bleeding follows.
- Technique (Fig. 94.11): **eviscerate small bowel → incise retroperitoneum over the aorta just above the inferior mesenteric artery** → extend up to the **ligament of Treitz** → follow the anterior aorta up to the **left renal vein crossing anteriorly** → vessel loop on the vein → **retract vein cephalad** (vein or Gil-Vernet retractor) → **loop both renal arteries** beneath. Right renal vein: via the same incision, or **reflect the second part of the duodenum.**
- **Haematoma obscuring the aorta: incise medial to the inferior mesenteric vein** → dissect through haematoma to the anterior aorta → follow cephalad to the left renal vein.
- Then: **incise peritoneum lateral to the colon** (release splenic or hepatic flexure attachments) → reflect colon off Gerota's fascia → **open Gerota's fascia** → dissect kidney from haematoma → **occlude pre-isolated vessels if bleeding.**

Damage Control

- Objective: **survival.** Patients show the **triad of hypothermia, metabolic acidosis and coagulopathy.**
- Sequence: **laparotomy to control haemorrhage and bowel contamination → temporary abdominal closure → ICU resuscitation and correction of coagulopathy → return for definitive surgery.** May improve renal salvage.
- **Oozing from Gerota's fascia → pack with laparotomy pads; return at 24–48 hours.**
- **Observe contained, non-pulsatile, non-expanding zone II haematomas** — evidence favours observation.
- Allows stabilisation before lengthy reconstruction, and may permit **AE instead of nephrectomy.** **Massive injury with persistent instability → quick damage-control nephrectomy** (lifesaving).
- Survival predictors in damage-control surgery: **Glasgow Coma Scale <8, base excess <-13.9, low preoperative diastolic pressure.**

Nephrectomy

- Decision depends on **overall status, reconstructability, whether the kidney causes the instability, associated injuries, a functioning contralateral kidney** (imaging, palpation or one-shot IVP) and team experience.
- **Unstable, life-threatening renal bleeding where damage control is not an option → total nephrectomy.**
- Reasons for nephrectomy: **most for extent of parenchymal/vascular injury; about a quarter for instability in an otherwise reconstructable kidney.**
- **Predictors: high Injury Severity Score, high renal grade, haemodynamic instability;** also penetrating trauma and laparotomy.
- Thresholds vary, but **nephrectomy rates are falling; urological involvement in the multidisciplinary team → higher salvage.**

Renal Reconstruction

- Principles, in sequence: **complete renal exposure** → **temporary vascular control** → **limited debridement of nonviable tissue** → **haemostasis by individual suture ligation** → **watertight collecting-system closure** → **reapproximate parenchymal defect** → **cover with Gerota's fascia or omentum** → **liberal drainage** (Fig. 94.12). **Preserve Gerota's fascia and perirenal fat during exposure** for later cover.
- **Renorrhaphy** (Fig. 94.13): debride → haemostasis and collecting-system closure → **approximate margins (3-0 absorbable) with capsule over a haemostatic bolster** (oxidised regenerated cellulose or gelatin sponge).
- **Unreconstructable polar injury** → **partial nephrectomy**; cover raw surface with a **pedicled omental flap** (rich blood and lymphatic supply → less delayed bleeding and leak); alternatives **polyglactin mesh, peritoneal graft, retroperitoneal fat**.
- **Gelatin-thrombin haemostatic sealants** are potent and increasingly used (less blood loss than suture alone in an animal model).
- Associated injuries are common (**liver and spleen most**; colon, pancreas, stomach). Previously nephrectomy was advised; **renal repair alongside bowel injury is now successful with fewer complications than conservative management**.
- NTDB: **surgical repair succeeds more often in grade IV than grade V**; failures mostly salvaged by nephrectomy. **AE achieves higher renal salvage than surgical repair**.

Renovascular injuries

- Uncommon; usually with associated injuries needing surgery. **Major renovascular injury with two kidneys** → **prompt nephrectomy**.
- **Absolute indications for attempted revascularisation: solitary functioning kidney or bilateral renal artery injury**.
- Salvage depends on **warm ischaemia time, ideally <120 minutes** (some success beyond). Even in expert hands, **main renal artery reconstruction salvages only about a third**.
- Repair: vascular clamps; **5-0 non-absorbable vascular suture** (Fig. 94.15: segmental veins may be ligated safely).
- **Blunt main renal artery thrombosis** — sequence: **deceleration** → **renal mobility stretches the artery** → **intima (poor in elastic fibres) tears** → **thrombus** → **ischaemic kidney** (Fig. 94.16: absent perfusion on CT, occlusion on arteriography).
- **Unilateral thrombosis: revascularisation rarely succeeds** (few successes, most with reduced function) → **observe if contralateral kidney normal**. **Bilateral hilar injury** → **attempt revascularisation** — up to about half avoid dialysis.
- Endovascular stenting for intimal-flap thrombosis is promising; the main drawback is **post-stent anticoagulation in polytrauma**.
- After repair of thrombosis, **hypertension develops in at least 43%** → **monitor blood pressure for up to 1 year**.
- **Penetrating hilar injury**: right-sided injury often involves **pancreaticoduodenal complex, renal vessels and inferior vena cava** → usually nephrectomy. If stable, repair with **5-0 or 6-0 polypropylene**; partial venous occlusion ideal.
- **Left renal vein can be ligated proximal to gonadal and adrenal branches** (collateralised); **right renal vein cannot** (no collaterals). **Artery: simplest repair possible** to limit ischaemia.
- **Renal vein injuries**: limited literature; conservative management and selected repairs without new hypertension, renal impairment or nephrectomy.

RENAL ARTERY INJURY — UNILATERAL VERSUS BILATERAL OR SOLITARY KIDNEY

Unilateral, normal contralateral kidney	Bilateral injury or solitary kidney
Observe (thrombosis) or nephrectomy (major vascular injury); revascularisation rarely succeeds	Attempt revascularisation — absolute indication; aim for warm ischaemia <120 min

Concept — *The kidney tolerates warm ischaemia poorly, so revascularisation is worth its risk only when the alternative is dialysis; with a normal contralateral kidney, observation or nephrectomy wins.*

Complications

- Priority: **survival first, renal salvage second**. Complications depend on patient, mechanism, grade, associated injuries and management.

Mortality

- Death reflects **the trauma and associated injuries**, not the renal injury itself. High-grade renal trauma mortality about 5–8%.
- **Nephrectomy and mortality — conflicting evidence**: MiGUTS and NTDB found **nephrectomy an independent predictor of death**; a propensity-adjusted TQIP analysis found **no independent association**.

Acute kidney injury

- **Acute kidney injury (AKI) is multifactorial** — injury severity, management, polytrauma and baseline factors.
- Commoner after nephrectomy, but on multivariable analysis **nephrectomy was not associated with AKI**; **age, male sex and transfusion requirement** were.
- **Functional loss on dimercaptosuccinic acid (DMSA) scan correlates with AAST grade**. Long-term renal function is poorly documented.

Persistent urinary extravasation, urinoma and perinephric abscess

- Persistent leak → **urinoma, perinephric infection, rarely renal loss**.
- Stepwise drainage: **internal stent ± urethral catheter** → **add percutaneous nephrostomy or percutaneous urinoma drain** if not cured.
- **Perinephric abscess is rare; persistent extravasation and urinoma are its precursors**. Management: **stent ± nephrostomy then percutaneous abscess drainage**; open surgery rarely needed.
- **Devitalised parenchyma, especially with intra-abdominal injury** → more morbidity, abscess and persistent leak.

Delayed bleeding

- May occur **up to several weeks** after injury; **usually within 21 days**.
- Initial management **bed rest and hydration** → persistent bleeding → **angiography and embolisation**.

Hypertension

- **De novo hypertension appears uncommon**; reported rates vary widely, true incidence unknown; may be early or late. Children: uncommon. Incidence tracks the **background prevalence of essential hypertension**.
- **Higher after renal artery injury or occlusion (about a third to a half)**. In one series, new hypertension was **associated with high-grade injury**.
- **No correlation between management choice** (nephrectomy, AE) and later hypertension.
- Mechanism common to all: **partial renal ischaemia → renin-angiotensin activation**.

POST-TRAUMATIC HYPERTENSION — MECHANISMS

Eponym / entity	Distinguishing mechanism	Management
Goldblatt kidney	Stenosis or occlusion of main renal artery or branch	—
Page kidney	Extrinsic parenchymal compression by blood or urine, or haematoma scarring	Acute: medical — angiotensin-converting enzyme inhibitor or angiotensin receptor blocker while haematoma resorbs. Refractory hypertension or renal dysfunction: percutaneous drainage or open/laparoscopic subcapsular decortication
Post-traumatic AVF	Arteriovenous shunt causing distal ischaemia	—

Other complications

- Uncommon: **new hydronephrosis or UPJ obstruction** from fibrosis (without ureteral injury), **chronic flank pain, renocutaneous fistula**.

Follow-up after trauma

FOLLOW-UP RECOMMENDATIONS BY GUIDELINE BODY

Body	Recommendation
EAU	Advocates follow-up (weak evidence). Grade I–II: no imaging. Grade III–V salvaged, asymptomatic: DMSA to document function
AUA	No specific follow-up recommendation
Urological Society of Australia and New Zealand (USANZ)	Renal ultrasound at 4–6 weeks (except grade I); check microhaematuria resolved at 6 months

- **New flank pain, haematuria, fever, hypertension or renal dysfunction → repeat imaging and bloods.**
- Minimum: **periodic blood pressure, examination, urinalysis and renal function.**
- **Return to activity/sport: 2 weeks to 3 months by grade** (expert opinion). High-grade salvaged kidneys risk **secondary bleeding and pseudoaneurysm/AVF rupture, especially at 2–6 weeks**; some reimage or confirm haematuria resolution first.
- **Contact-sport risk to the kidney is very low; a functional solitary kidney after trauma does not preclude team or contact sport.**

KEY POINTS: RENAL TRAUMA

- **Expectant management maximises renal preservation.**
- **Degree of haematuria and severity of injury do not consistently correlate.**
- **Contrast-enhanced CT is the best imaging** for assessing and grading renal trauma.
- **Microhaematuria without shock → observe without imaging.**
- **Stable, well-staged injuries — even high grade — can be managed conservatively.**
- **Selective embolisation** stops active bleeding from parenchymal lacerations and segmental arterial injury.
- CT red flags: **medial haematoma** (pedicle), **medial extravasation** (pelvis/UPJ), **absent enhancement** (main artery), **active intravascular contrast extravasation** (brisk arterial bleeding).
- **Instability despite resuscitation → immediate intervention** (surgery, or AE in selected cases).

- **Urinary extravasation usually resolves conservatively; renal pelvis or proximal ureteral avulsion demands prompt intervention.**

Concept — Renal trauma management is governed by one asymmetry: haemodynamic stability buys time for staging and observation, and every modern gain in salvage — imaging, AE, damage control — comes from converting would-be explorations into observation.

URETERAL INJURIES

Epidemiology

- **Acute ureteral trauma is uncommon: 1–2.5% of genitourinary trauma.**
- External trauma: **penetrating predominates; blunt up to about 40%.**
- **Iatrogenic injury is the commoner cause overall** — open, endoscopic and minimally invasive surgery, mainly gynaecological and colorectal.
- Mechanisms: **suture ligation, incision/transection, avulsion, devascularisation, thermal energy** (microwave, electrocautery, ultrasonic) and **cryoablation**.
- **Often unrecognised until complications** — urinoma, abscess, sepsis, renal failure, stricture, fistula. **Delayed diagnosis after penetrating trauma → more nephrectomies and longer stay. Maintain a high index of suspicion.**

External Trauma

- Rare; classified by the **AAST ureter scale** (Table 94.3). **Young men; penetrating; gunshot wounds commonest** (up to ~90%). **Blunt injury: mostly motor vehicle collisions.**
- **Wartime: about 5% of urological injuries involve the ureter.**
- **>90% have a concurrent abdominal or retroperitoneal organ injury**, especially penetrating: **small bowel (commonest)**, large bowel, vascular; also kidney (10–28%), less often bladder. **Blunt ureteral injury → bony injuries.**
- **Proximal ureter is the commonest site of external injury**; mid and distal less so (blunt mid-ureteral injury very rare).
- **Mortality 6–9%**, reflecting associated injuries.
- **Ballistic mechanism: direct transection plus disruption of the intramural blood supply**; microvascular damage up to **2 cm** from the transection, yet ureters usually need only **minimal debridement to a bleeding edge**.
- **Blunt: extreme whole-body force** (fall, pedestrian, high-speed collision); associated with **lumbar transverse process fractures and thoracolumbar dislocation**. **Massive-force injury → suspect ureteral injury. Rapid deceleration was present in every UPJ injury** in one series.
- **Absence of haematuria does not exclude ureteral injury.**

TABLE 94.3 AMERICAN ASSOCIATION FOR THE SURGERY OF TRAUMA ORGAN INJURY SEVERITY SCALE FOR THE URETER

Grade	Type	Description
I	Haematoma	Contusion or haematoma without devascularisation
II	Laceration	<50% transection
III	Laceration	≥50% transection
IV	Laceration	Complete transection with <2 cm devascularisation
V	Laceration	Avulsion with >2 cm devascularisation

Advance one grade for bilateral injury, up to grade III. Data from Moore et al., 1992.

- **Pattern:** two cut-offs — **50% of circumference** (II vs III) and **2 cm of devascularisation** (IV vs V).
- **Trap:** grade IV is **complete transection**; grade V is **avulsion** — the distinction is **length of devascularisation**, not completeness.

Surgical Injury

- **Any abdominopelvic operation can injure the ureter. Distal ureter is the commonest site of iatrogenic injury** in non-urological surgery.
- Incidence by specialty: **colorectal ~0.2–0.4%**, **gynaecological ~0.3%**; vascular and spinal less.
- Laparoscopy (1960s) and robotics (1990s) initially **raised iatrogenic injury rates**, which **fell with experience**.
- Meta-analysis: **laparoscopic gynaecological surgery carries higher risk than open**; **laparoscopic and open colorectal surgery do not differ**.
- **Uteroscopy:** injury 0.13–9.5%; **avulsion <1%**; most minor injuries heal, **0.3–3% end in stricture**.
- **Thermal injury → ureter**; **lysis of adhesions → bladder**.
- Historical contributors: **hysterectomy about half**, then colorectal, pelvic (ovarian), transabdominal urethropexy, abdominal vascular surgery.
- **Medicolegal:** malpractice claims most often follow **gynaecological surgery**, then urological.

FIG. 94.17 — REPORTED INCIDENCE OF IATROGENIC URETERAL INJURY (EAU UROLOGICAL TRAUMA GUIDELINES)

Procedure	Reported incidence (%)
Gynaecological: emergency caesarean section · vaginal hysterectomy · abdominal hysterectomy	0.01–0.06 · 0.02–0.5 · 0.03–2.0
Gynaecological: minimally invasive hysterectomy · urogynaecological (prolapse)	0.13–6.0 · 1.7–4.3 (highest gynaecological)
Colorectal	0.15–10
Endourology: mucosal abrasion · perforation · intussusception/avulsion	0.3–4.1 · 0.2–2.0 · 0–0.3
Radical prostatectomy: open · robotic-assisted	0.05–16 · 0.05–0.4

- **What it tests:** within gynaecology, risk rises **vaginal → abdominal → minimally invasive → prolapse surgery**; **robotic prostatectomy has a far narrower range than open**; in endourology, **avulsion is the rarest injury**.

Obstetric and gynaecological surgery

- Rare but morbid, especially if unrecognised. **Repeat caesarean section** accounts for a notable share; **placenta accreta spectrum, especially percreta**, raises urinary tract injury risk. Historically, **tubal ligation** caused many injuries.
- Laparoscopic hysterectomy injury depended heavily on **surgeon experience** during the learning curve. **Robotic and laparoscopic hysterectomy rates are now equivalent (~0.9%)**; **minimally invasive hysterectomy carries higher risk than open abdominal**.
- National Surgical Quality Improvement Program (NSQIP) benign hysterectomy: ureteral obstruction ~0.1%, fistula ~0.05%; **mean diagnosis at 10 days, rarely on the day of surgery**. **Delay → readmission, complications, sepsis, AKI**.
- **Minimally invasive sacrocolpopexy: very low rate**.
- **Endometriosis raises risk**, for three reasons: **extrinsic or intrinsic ureteral involvement**; **adhesions obscuring the ureter**; **medial deviation of the ureter**.

Colon and rectal surgery

- Rare; laparoscopic colectomy once thought riskier, but meta-analysis shows no difference from open.

Urological surgery

- Open urological procedures were historically a significant source. **First ureteroscopic injury reported 1984**; injuries surged with ureteroscopy in the late 1980s, and **ureteroscopy is the commonest iatrogenic cause in some modern series**.
- **Post-ureteroscopy stricture rate is stable or falling (~3%)**. After ureteroscopic treatment of **upper tract urothelial carcinoma**, stricture is the commonest complication; perforation 1.3–7.4%.
- **Continued basketing after a recognised tear causes injury → stop and place a stent** when perforation is identified. **Holmium:yttrium-aluminium-garnet (Ho:YAG) fragmentation before basketing** reduces this. **Extruded stone fragments are minor** and rarely cause stricture.
- **Work alongside or over a safety wire into the renal pelvis** (facilitates stenting); some experts omit it in routine flexible ureteroscopy.
- **Risk factors for ureteroscopic complications**: long operating time, **12/14-Fr access sheath for renal stones**, **impacted stone**, **proximal location**, surgeon inexperience, **previous irradiation**, stone size and number, congenital anomalies.
- Lithotripsy injury risk, highest first: **electrohydraulic lithotripsy (EHL, largely historical) → neodymium:YAG (Nd:YAG) → Ho:YAG**.
- Protective: **smaller, flexible ureteroscopes**. Access sheaths protect the ureter but can injure its wall, especially if not pre-stented → use **≤14-Fr sheaths in unstented ureters**.
- **Robotic prostatectomy: ureteral injury <1%**.
- **Partial nephrectomy or thermal ablation → proximal ureteral stricture** (rare, ~0.35–1.5% open); presents **months later** (earlier after ablation than after partial nephrectomy); linked to **lower-pole and central/hilar tumours** compromising proximal ureteral blood supply. **Pyeloperfusion during ablation** protects the ureter near high-risk tumours.

Vascular surgery

- Rarely injures the ureter. **Transient, asymptomatic hydronephrosis after aorto-iliac/aortofemoral bypass** (12–20%) resolves spontaneously; **symptomatic stenosis from devascularisation or inflammation in 1–2%**, often **months later**.
- Risk factors: **reoperation**, **periaortic fibrosis**, **inflammatory large aneurysm**, **graft placed anterior to the ureter**.
- **Up to 85% not recognised immediately**; present with **flank pain, fever, ileus, distension, urinary fistula**.
- **Ureteroarterial fistula**: rare, potentially catastrophic. **Commonest presentation significant haematuria** (may be intermittent or microscopic). Usually **ureter to ipsilateral iliac artery** (sometimes aorta). Associated with **chronic indwelling stent (~80%; mean ~3 years)** plus pelvic surgery or radiation, pelvic malignancy, vascular disease, infection or pregnancy. **Treatment: endovascular covered stent bridging the fistula**. Mortality has fallen to **~7% since 2000**.

Tenuous ureteral blood supply

- **Distal ureteral supply is variable**: in **~10% of women**, a disproportionate share comes from **uterine artery branches**, necessarily divided at hysterectomy; **~40% show reduced ureteral perfusion after uterine artery ligation** (cadaveric).
- Hence **distal devascularisation may be unavoidable** in a few hysterectomies.
- **Devascularisation presents late** (>1 week, up to 1–2 months) as **stenosis, urinoma or ureterovaginal fistula** not evident at surgery.

Avoiding and detecting ureteral injury

- **Know the ureter's relation to the uterine and ovarian arteries** before ligation (Fig. 94.18).
- **Open surgery: up to a third recognised immediately; laparoscopy: fewer** → high index of suspicion. Post-operative signs: **fever, abdominal pain, port-site pain, leucocytosis**; fewer present with haematuria or a pelvic mass (urinoma). **Low threshold for imaging.**
- **Uterosacral ligament region** is hardest to visualise. **Uncontrolled bleeding raises risk** — haemostasis and exposure reduce it.
- Hydration or diuretics to aid visualisation: **no supporting data.**
- **Intraoperative recognition and repair** → better success, less morbidity.

Preoperative ureteral stenting

- **Aids identification in high-risk cases** and **improves intraoperative recognition** of injury (avoiding delayed diagnosis).
- **Whether stents reduce injury is conflicting:** gynaecological surgery — possible reduction, **none in laparoscopic surgery**; colorectal — older studies and meta-analysis no benefit, a NSQIP analysis found fewer injuries after colectomy.
- **Lighted ureteral catheters;** a **small 5-Fr catheter** may reduce oedema and obstruction seen with larger catheters.
- **Intraureteral indocyanine green (ICG) with near-infrared fluorescence:** open and minimally invasive, gynaecological and colorectal; **5 mL of 2.5 mg/mL per ureter** fluoresces for hours (median ~8 hours) (Fig. 94.19).
- **Stent complications:** anuria after bilateral prophylactic stents (1–5%), iatrogenic ureteral injury ~1%, urinary tract infection (UTI), AKI, longer operating time. **Guidewire use reduces serious (Clavien IIIb) complications.**
- Cancer-centre series of perioperative catheters: mostly **minor (Clavien I–II) complications** — **urinary retention, transient AKI, UTI**; a minority needed post-operative stent or nephrostomy, often after **difficult placement.**

Intraoperative patency checks

- **Routine retroperitoneal opening around hysterectomy has been abandoned** — it can devascularise the distal ureter.
- **Palpation through the unopened retroperitoneum, or pinching to provoke peristalsis, is unreliable.**
- **Cystoscopy alone** (absence of haematuria, bilateral jets) is traditionally a **poor predictor** with variable detection, but **remains recommended and cost-effective at minimally invasive hysterectomy and prolapse surgery.**
- Enhance efflux with **IV methylene blue, indigo carmine, fluorescein, or oral phenazopyridine (most cost-effective);** or **mannitol or 10–50% dextrose irrigation.**

AGENTS TO VISUALISE URETERAL EFFLUX OR THE URETER

Agent	Route	Key features / cautions
Methylene blue	IV	Potent monoamine oxidase inhibitor (MAOI) → serotonin toxicity with selective serotonin reuptake inhibitors (SSRIs), serotonin-noradrenaline reuptake inhibitors (SNRIs), MAOIs. Avoid in pregnancy and glucose-6-phosphate dehydrogenase (G6PD) deficiency (methaemoglobinaemia, haemolysis). Falsely lowers pulse oximetry
Indigo carmine	IV	Safe; rare hypotension (commonest) , bronchospasm, bradycardia, hypertension, anaphylactoid reaction. Half-life 4–5 min; blue urine in 5–10 min

Agent	Route	Key features / cautions
Phenazopyridine	Oral, preoperative	Most cost-effective
Fluorescein	IV	Alternative urine dye
Indocyanine green	Intraureteral via catheter	Near-infrared fluorescence of the ureter itself; prolonged (hours)
Mannitol or 10–50% dextrose	Bladder irrigation	Improves visualisation of efflux

Concept — *Iatrogenic injury is a problem of recognition more than of repair: minimally invasive surgery, thermal energy and devascularisation all produce injuries that are invisible at the time and declare themselves days to weeks later.*

Diagnosis

- Penetrating trauma with any haematuria, or a suggestive injury pattern → CT abdomen/pelvis with urogram.
- Blunt trauma with gross haematuria, microhaematuria with hypotension, significant deceleration, or major associated injuries → image.
- Haematuria is nonspecific; 25–45% of ureteral injuries have no haematuria → high index of suspicion after penetrating trauma.

Intraoperative recognition

- After violent injury, **most injuries are recognised promptly, over half intraoperatively**. Use **injury pattern and trajectory**; in emergency exploration without imaging, **evaluate the ureter if haematuria or a suggestive mechanism**. Direct retroperitoneal exploration may be hampered by hematoma.
- **Iatrogenic intraoperative consultation**: establish the procedure and its status, haemodynamic and anaesthetic state, review imaging and history; **change approach or position if needed**; evaluate contralateral ureter and bladder. Tools: **direct inspection, cystourethroscopy** (efflux, wire or stent ± fluoroscopy), **IV methylene blue or indigo carmine**.
- **Avoid devascularising the ureter** during exploration. Alternatives: **cystoscopic ureteral catheterisation, or cannulation of the orifices via cystotomy with retrograde methylene blue**.
- **Missed injury** results from **inadequate exploration or low suspicion amid multiple injuries**: ~11% miss rate at laparotomy across trauma centres; **most injuries after hysterectomy missed** in one study. **Delay → longer stay, more nephrectomy, death**; also urinoma, abscess, stricture, fistula.
- **Fever, leucocytosis and local peritoneal irritation** are the commonest signs of a missed injury → CT. **Retrograde ureterography** is the usual test for missed injury in a stable patient — **permits immediate stenting**.

Imaging Studies

IMAGING FOR SUSPECTED URETERAL INJURY

Modality	Role	Key points
CT urography with delays	Best study in stable patients (AUA)	Delayed images at 10 min essential; trace both ureters entire length
Retrograde ureterography	Most sensitive test	Shows site and extravasation; time-consuming; limited acute role if unstable; mandatory if suspicion persists with non-diagnostic urogram ; allows simultaneous stenting — main use for missed injuries

Modality	Role	Key points
Antegrade ureterography	Retrograde not possible (e.g. wide gap)	Follow with antegrade stent or nephrostomy
IVP	Resource-poor settings	One-shot IVP non-diagnostic in 33–100% — never rely on it alone

- **CT signs: no contrast in the ureter on delays, delayed or asymmetric ipsilateral nephrogram, contrast extravasation, ipsilateral hydronephrosis, no contrast in the distal ureter, periureteral urinoma, asymmetric excretion.**
- Modern CT scans **before contrast is excreted** — hence the mandatory delay.
- **Significant UPJ laceration: medial extravasation or non-opacified ipsilateral ureter, or circumrenal extravasation. Medial or circumrenal extravasation with non-opacified distal ureter → avulsion** (Fig. 94.20: absent contrast below the injury indicates complete transection).
- IVP signs: **incomplete ureteral visualisation, deviation or dilatation, extravasation, hydronephrosis.**
- Diagnostic insensitivity and false negatives → **delayed detection in 8–20%.**
- Fig. 94.21: flank gunshot wound with incidental congenital UPJ obstruction — retrograde study showed an intact ureter; nephrostomy and antegrade study **localised the injury to the renal pelvis**, illustrating the need to combine studies.

Concept — *Ureteral injury is diagnosed by the absence of something — contrast beyond the injury — so the study must be timed for excretion, both ureters must be followed to the bladder, and a non-diagnostic study in the face of suspicion is not a negative one.*

Management

AUA Urotrauma Guideline:

- **Inspect the ureters directly at laparotomy** if injury is suspected without preoperative imaging.
- **Stable patient: repair traumatic lacerations at the time of laparotomy.**
- **Unstable patient: temporary urinary drainage, delayed definitive repair.**
- **Contusion at laparotomy: stent, or resect and repair**, according to viability and scenario.
- **Incomplete injury diagnosed post-operatively or late: attempt stent**; if unsuccessful or impossible, percutaneous nephrostomy with delayed repair.
- **Ureterovaginal fistula: stent first**; surgery if stenting fails.

Endoscopic management of ureteral injuries

AUA Urotrauma Guideline:

- **Endoscopic injury → stent and/or nephrostomy. Inadequate diversion → periureteral drain. Delayed reconstruction often required. Open repair** if endoscopic or percutaneous drainage fails or is impossible.
- Endoscopic stenting is appropriate for **partial injuries not otherwise undergoing laparotomy** and for **delayed recognition.**
- Blunt ureteral injury (NTDB): **most managed by stent and/or nephrostomy**; reconstruction in a substantial minority of grade III–V, **more likely if laparotomy was performed.**

General principles of ureteral reconstruction

- **Repair at the initial laparotomy in a stable patient; not in the unstable polytrauma patient.** Ureteral blood supply is tenuous; imperfect repair → **leak, debility, nephrectomy, rarely death.**

- **Unstable or repair impossible:** ligate the ureter with long silk or polypropylene and repair later, or place nephrostomy after ICU resuscitation (damage control); or temporary cutaneous ureterostomy over a single-J stent or paediatric feeding tube, tied proximally to secure the tube and stop leakage.

1. Mobilise carefully, sparing the adventitia.
2. Debride minimally but judiciously to bleeding edges, especially high-velocity gunshot wounds.
3. Spatulated, tension-free, stented, watertight anastomosis with fine absorbable suture; retroperitoneal drain.
4. Retroperitonealise the repair where possible.
5. Do not tunnel ureteroneocystostomy — wide, spatulated, non-tunnelled anastomosis.
6. Omental interposition for severe injury, blast effect, concomitant vascular repair or complex cases.
7. No bowel interposition or autotransplantation acutely.

FIG. 94.22 — REPAIR OPTIONS BY LEVEL OF URETERAL INJURY

Level	Acute repair	Delayed / stricture / segmental loss
Upper	Ureteropyelostomy; ureteroureterostomy; ureterocalicostomy (rare)	Buccal graft ureteroplasty; appendiceal interposition (right); ileal or Yang-Monti interposition
Mid	Ureteroureterostomy (± downward nephropexy); Boari flap/psoas hitch	Transureteroureterostomy (TUU); ileal interposition; buccal ureteroplasty
Distal	Ureteroneocystostomy ± psoas hitch	Boari flap

External trauma — damage control

- Defer definitive repair when the patient cannot tolerate the operating time: severe haemorrhagic shock, uncontrollable bleeding, severe colonic injury (especially colectomy) → staged repair after full resuscitation.
- Four damage-control options: (1) do nothing, re-operate within ~24 hours; (2) stent into proximal ureter, exteriorised to skin; (3) cutaneous ureterostomy; (4) ligate and plan staged percutaneous nephrostomy.
- Authors' preference: ligate the proximal stump with long silk or polypropylene (aids later identification) → percutaneous nephrostomy once fully stabilised, not intraoperatively; open nephrostomy is too slow — no role in the unstable patient.
- Alternative: single-J stent or 5-Fr feeding tube into the proximal ureter with minimal mobilisation, secured with a non-absorbable tie, externalised into a urostomy bag.
- Definitive reconstruction after functional and anatomical imaging.

Contusion

- Most minor injury, but microvascular damage → necrosis, stricture or breakdown; common with close-range gunshot blast.
- Severe or non-viable contusion → excise and perform ureteroureterostomy or ureteroneocystostomy. Intact contused ureter at laparotomy → stent (cystoscopy or cystotomy).
- Missed contusion presenting as leak: retrograde imaging and stent; if impossible, nephrostomy and antegrade stent for 6 weeks (good success, ~85%).

Upper ureteral injuries

AUA: **proximal to the iliac vessels** → **primary repair over a stent; no bowel interposition or autotransplantation acutely.**

- **Avulsion from the renal pelvis or very proximal ureter** → **reimplant ureter into the renal pelvis** (spatulated; Fig. 94.23) — open, or delayed laparoscopic/robotic. Difficult because the **intrarenal pelvis is often non-dilated.**
- Unstable: **nephrostomy**, plus **periureteral drain** if leak persists. Stable: **immediate repair** may be considered.
- **Ureteroureterostomy over a stent** (end-to-end; Figs 94.24–94.25): for **upper two-thirds, proximal to the iliac vessels**; commonest acute repair; **up to 90% success.** Technique: **define injury by mobilisation** → **debride and spatulate** → **stent** → **approximate with 5-0 absorbable suture.**
- Complications: **urine leak (10–24%)**, periureteral abscess or urinoma, **stricture (up to ~12%).** Long-term outcome unknown (poor trauma follow-up).
- **Prolonged drain output after UPJ repair** is common yet benign (mean ~12 days in one series); **retroperitonealisation may shorten leakage.**
- **Ureterocalicostomy** (ureteral stump end-to-side to an exposed lower calyx): for **profound pelvis/UPJ damage**; requires **lower-pole partial nephrectomy**; technically hard (small medial ureter to large lateral calyx).
- **Robotic surgery** for delayed reconstruction in stable patients: pyeloplasty, ureteroureterostomy, ureterocalicostomy. **Long delayed strictures** → **robotic buccal graft ureteroplasty** (~87% intermediate success).
- **Nephrectomy is rarely needed.** Acute: **instability with severe visceral injury** (damage control nearly always preferable) or **unreconstructable ipsilateral renal injury.** Delayed: **poor function** (late-recognised obstruction), **pan-ureteral injury with reconstruction impossible, persistent (especially vascular) fistula.** **Avoid nephrectomy whenever possible.**

Midureteral injuries

- Mostly **complete transections above the iliac vessels** (Figs 94.26–94.27: suture ligation; transection after anterior spinal surgery, complicated by pseudoaneurysm after nephrostomy requiring embolisation) → **primary ureteroureterostomy over a stent.**
- Defect too long for tension-free repair → **downward nephropexy** (whole-kidney mobilisation; classically **more length on the left** — longer vessels; up to 7 cm reported, a few centimetres typical and similar on both sides) or **TUU.**

Transureteroureterostomy

- **Rarely used but 90–97% successful** (paediatric ~70%). Injured ureter crosses the midline → **end-to-side into contralateral ureter** via a **≥2 cm recipient ureterotomy.**
- **Little role acutely** — typically secondary/delayed; acutely use **ligation or cutaneous diversion over a stent.**
- Indicated when **ureteroureterostomy or bladder flap/hitch impossible: severe bladder scarring, small bladder, very long ureteral loss, pelvic phlegmon, or associated rectal, major pelvic vascular or extensive bladder injury.** Laparoscopic TUU reported in children.
- **Relative contraindications:** urothelial carcinoma, nephrolithiasis, genitourinary tuberculosis, pelvic radiation, retroperitoneal fibrosis, chronic pyelonephritis — seldom known at trauma laparotomy.
- **Risk:** converting a unilateral into an iatrogenic bilateral injury.

Lower ureteral injuries

AUA: **distal to the iliac vessels** → **reimplantation or primary repair over a stent**; may need **psaos hitch or Boari flap**.

- **Ureteroneocystostomy**: injury close enough to the bladder that **no hitch or flap is needed**. Principles: **long, non-tunnelled, spatulated, stented**.
- **Refluxing anastomoses** (neobladder, ileal conduit) show **no excess reflux complications**; in neobladders, **non-refluxing anastomoses** **stricture more** with renal deterioration → **non-tunnelled preferred in trauma**.
- Elective, non-acute: **robotic ureteroneocystostomy ~95% success**; novel **non-transecting side-to-side reimplant** preserves ureteral vascularity.

URETERONEOCYSTOSTOMY IN TRAUMA — NON-TUNNELLED (REFLUXING) VERSUS TUNNELLED (NON-REFLUXING)

Non-tunnelled, refluxing — preferred	Tunnelled, non-refluxing
Very low risk of clinically significant reflux	Higher risk of stricture/obstruction and renal deterioration

Psoas bladder hitch

- **Mainstay for lower-third injury; 95–100% success** (Fig. 94.28). Preferred over ureteroureterostomy in the lower ureter because the **tenuous distal supply may not survive transection** (some prefer end-to-end if the distal stump is preserved).
- Technique: **suture bladder apex to ipsilateral psaos muscle and psaos minor tendon**; often **divide contralateral superior vesical pedicle**; **protect contralateral ureter**.
- **Place sutures along the length of the psaos** to avoid **genitofemoral nerve entrapment**; **avoid deep bites** → **femoral nerve injury**.
- Quick, but **only in the stable patient**; unstable → staged repair.

Boari flap

- **Lower two-thirds injuries with gaps too long for a hitch** → Boari flap or TUU (Fig. 94.29).
- Requires **normal bladder size and wall**: **inverted-U anterior wall flap**, base near the dome, **≥4 cm wide, length:width ≤3:1**; combined with **psaos hitch**; **bridges gaps up to 12 cm**.
- **Time-consuming — not for most acute injuries**; high success open or robotic (robotic series ~88%, failures within 3 months).

BRIDGING A DISTAL URETERAL GAP — HITCH VERSUS FLAP

Psoas hitch	Boari flap (+ hitch)
Lower third; shorter gaps; quick; 95–100% success	Lower two-thirds; gaps up to 12 cm ; time-consuming; needs normal bladder

Partial transection

- **Primary repair is used for most ureteral injuries**: spatulated, watertight, **interrupted or running 4-0/5-0 absorbable** (polyglactin, polydioxanone, poliglecaprone).
- **Close a longitudinal laceration transversely (Heineke-Mikulicz)** so as not to narrow the lumen; retroperitonealise. **Stent and retroperitoneal drain always**.

Monitoring after ureteral repair

- Little consensus. Typically **diuretic renogram or ultrasound with Doppler jets at 4–12 weeks and at 1 year**.
- After ureteroneocystostomy ± hitch/flap, **recurrence is rare** and routine imaging **beyond the first scan at 2–3 months detected no silent obstruction** — recurrences were symptomatic.

- Polytrauma needing CT for other injuries → **contrast CT with delayed images**.
- Because repairs are often in devascularised ureter, **late stenosis can occur and may be clinically silent**.

Delayed management of ureteral injury related to external trauma

- Depends on **location, mechanism, prior management, clinical course and functional status**; follows the same level-based paradigm. **Bowel interposition and autotransplantation become options only in the delayed setting**.
- **Autotransplantation**: after **profound ureteral loss or repeated failed repairs**; **no role acutely**; usually the **last resort before nephrectomy**. **5–10% of renal units lost** (lower in experienced transplant centres); early major complications (Clavien ≥III: bleeding, infection) ~15%; robotic autotransplantation emerging.
- **Ileal ureter (“chimney”)**: for **long-segment loss**; **81–100% success**. Long-term: anastomotic stricture and fistula uncommon; **about a fifth need reoperation**; **metabolic acidosis (~20%)**, recurrent pyelonephritis with refluxing anastomoses into reservoirs.
- **Yang–Monti**: short bowel segments reconfigured into a **long narrow tube** → **less bowel, less absorption** → fewer metabolic complications in **impaired renal function**. Laparoscopic and robotic ileal interposition described.
- **Appendix**: segmental replacement or onlay (open or minimally invasive; right side), ~92% short-term success.
- Most create a **wide-open, refluxing, non-tapered ileal ureter** (Fig. 94.30) — reflux is not clinically significant. Authors **combine with ipsilateral psoas hitch** to **shorten the conduit**, theoretically reducing metabolic and mucus complications.
- **Ileal interposition has no role in acute repair** — delayed or staged only.

ACUTE VERSUS DELAYED URETERAL RECONSTRUCTION

Acute (at laparotomy)	Delayed / staged
Stable: primary repair over stent — ureteroureterostomy, pyelostomy, ureteroneocystostomy ± hitch. Unstable: damage control (ligation + nephrostomy, exteriorised stent, cutaneous ureterostomy)	Boari flap, TUU , ileal/Yang–Monti or appendiceal interposition , buccal graft ureteroplasty , autotransplantation (last resort before nephrectomy)

Concept — *Level decides the operation and stability decides the timing: simple, stented, tension-free repairs are done at the first laparotomy, and anything needing bowel, a transplant or the contralateral ureter waits until the patient and the tissues have declared themselves.*